

SERVICE MANUAL

Invivo Corporation
3545 SW 47th Avenue
Gainesville, FL 32608

Document TR0223S

Revision 12

GE 1.5T 8 Channel High Resolution Brain Array

GE Part Number: M3087JA

Invivo Part Number: 800223



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	- Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod. - Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik. - Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.
VÝSTRAHA (CS)	Tento provozní návod existuje pouze v anglickém jazyce.
	- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. - Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. - V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	Denne servicemanager findes kun på engelsk.
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WAARSCHUWING (NL)	Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.
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WARNING (EN)	This service manual is available in English only.
	- If a customer's service provider requires a language other than english, it is the customer's responsibility to provide translation services. - Do not attempt to service the equipment unless this service manual has been consulted and is understood.

	Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.
HOIATUS (ET)	See teenindusjuhend on saadaval ainult inglise keeles - Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. - Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. - Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VAROITUS (FI)	Tämä huolto-ohje on saatavilla vain englanniksi. - Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. - Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. - Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
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ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά. - Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης. - Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις. - Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.
FIGYELMEZTETÉS (HU)	Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. - Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészíttetése. - Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. - Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.
AÐVÖRUN (IS)	Þessi þjónustuhandbók er aðeins fánleg á ensku. - Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu. - Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin. - Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.
Avvertenza (IT)	Il presente manuale di manutenzione è disponibile soltanto in lingua inglese. - Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. - Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. - Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla

manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

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BRĪDINĀJUMS (LV)	Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā. - Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu. - Neveiciet aprīkojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas. - Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.
ĮSPĖJIMAS (LT)	Šis eksploatavimo vadovas yra tik anglų kalba. - Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas. - Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo. - Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.
ADVARSEL (NO)	Denne servicehåndboken finnes bare på engelsk. - Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse. - Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. - Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. - Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. - Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go. - Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.
AVISO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. - Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. - Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. - A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.
ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. - Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. - Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. - O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.
ATENȚIE	Acest manual de service este disponibil doar în limba engleză.

(RO)	• Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. • Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. • Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. • Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. • Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. • Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.
UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. • Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge. • Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. • Zanimanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
upozornenie (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. • Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. • Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu. • Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCION (ES)	Este manual de servicio sólo existe en inglés. • Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. • No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. • La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
Varning (SV)	Den här servicehandboken finns bara tillgänglig på engelska. . • Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster. • Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. • Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
OPOZORILO (SL)	Ta servisni priročnik je na voljo samo v angleškem jeziku. • Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka zagotoviti prevod. • Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli. • Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevarnosti poškoduje ponudnik storitev, operater ali bolnik.
DİKKAT (TR)	Bu servis kılavuzunun sadece ingilizcesi mevcuttur. • Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. • Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. • Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

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SECTION 1 – INTRODUCTION

1-1 Product Identification and Shipping List

This is a service manual for the GE SIGNA 1.5T 8 Channel High Resolution Brain Array.



SHIPPING LIST – TABLE 1-1

Description	GE Part #	INVIVO #	Qty
1.5T 8 Ch HiRes Brain Array / Baseplate Assm.	2317112-2	101844	1
Phantom Positioner	2317112-3	102558	1
Operator Manual	2317112-4	500077	1
Service Manual	2317112-5	TR0223S	1
Thick Head Pad	2317112-9	102985	1
Thin Head Pad	2317112-10	102984	1
Spacer Pad	2317112-7	101622	1
Wedge Pad	2317115-8	101407	2

1-2 Compatibility

This coil is compatible with the following hardware configurations

- Signa Horizon and LX Horizon 1.5T with Excite Technology (8-channel)

1-3 Related Documentation

- Signa LX Service Methods CD, 2160623-1
- 8 Channel High Resolution Brain Array Operator's Manual, 2383610-4

1-4 Environmental Requirements

Storage Requirements

Coil should be stored flat in the scanner room.

Dimensions

Coil Dimension:	417.6 mm x 393.4 mm x 406.4 mm	(16.44 in x 15.49 in x 16.00 in)
Phantom Positioner:	225.6 mm x 228.6 mm x 23.9 mm	(8.88 in x 9.00 in x 0.94 in)

Weight

Coil Weight:	7.20 Kg	(16.0 lb)
Phantom Positioner:	0.45 Kg	(1.0 lb)

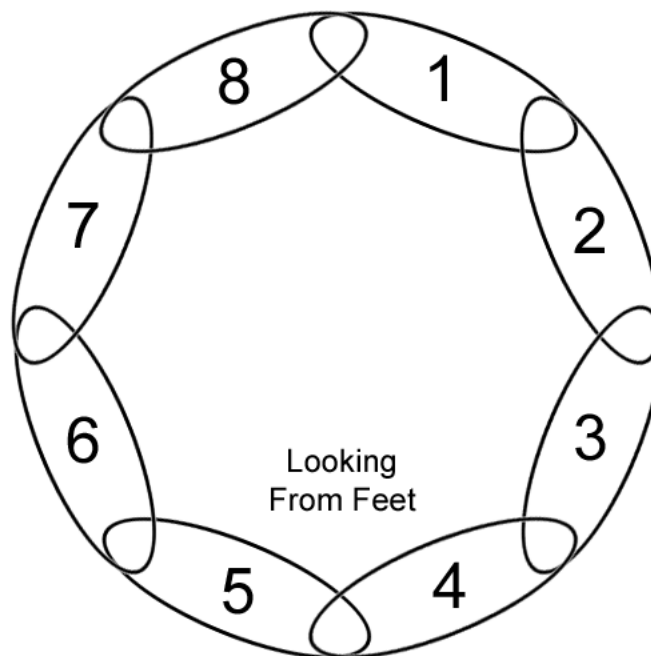
1-5 Theory of Operation

Refer to Figure 1 below and the Block Diagram on the following page.

The 1.5T 8 Channel High Resolution Brain Array consists of 8 elements. Each element is comprised of a loop arranged azimuthally around the head. Each loop is approximately 45° in azimuthal extent. The center of Loop #1 is located 22.5° rotated in the clockwise direction (as viewed from the feet) with respect to the nose. Each of the next seven loops is rotated clockwise by 45° successively. Loops #1 and #8 are located anteriorly while loops #4 and #5 are located posteriorly. Loops #2 and #3 are located on the patient Left side while Loops #6 and #7 are located on the patient Right side. During transmit, each loop is decoupled with both passive and active decoupling, with an additional active preamp protection network. During receive, signals from 1 – 8 are directed to separate ultra-low-input impedance preamp modules placed directly at the output of the loop, providing additional element decoupling.

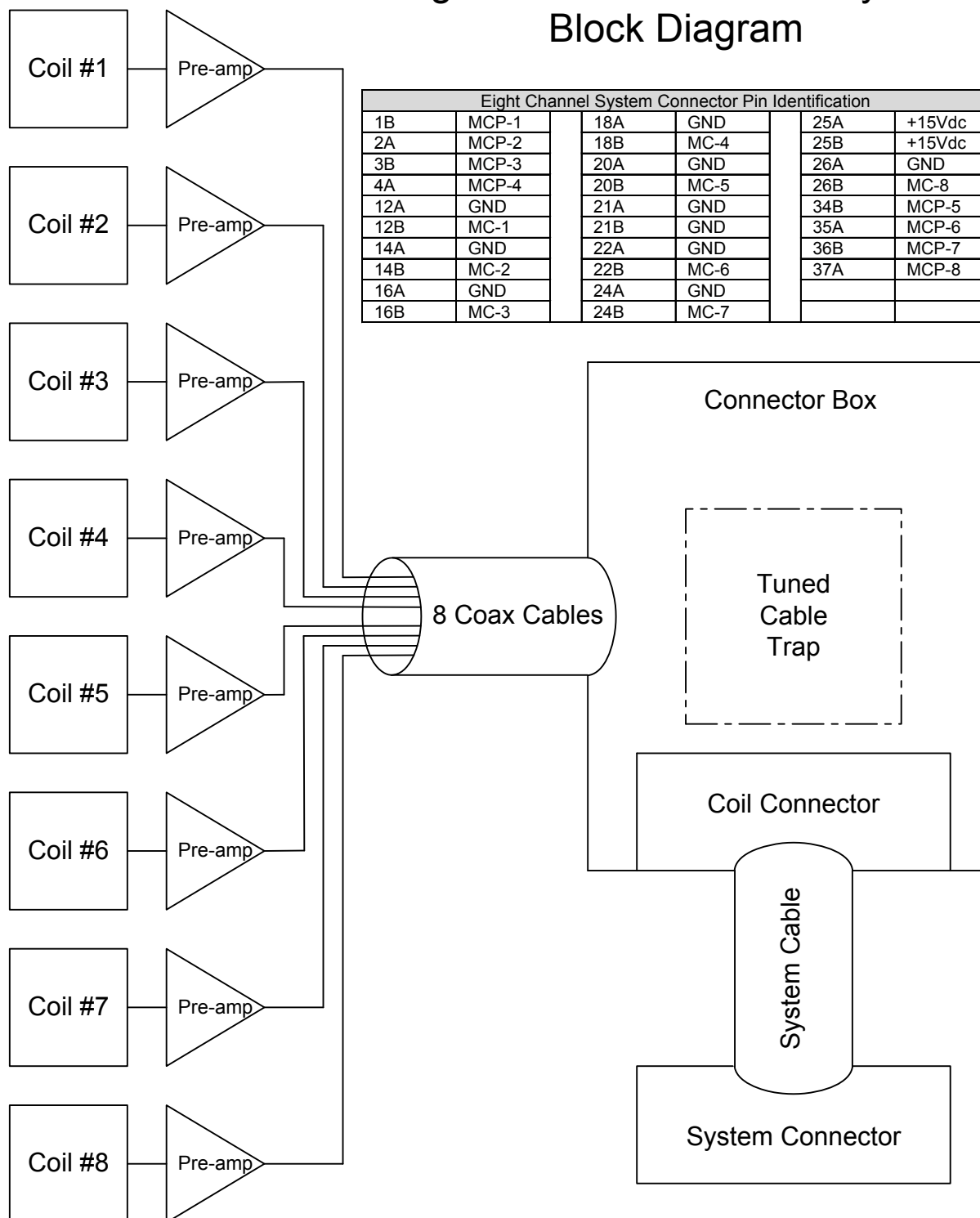
Figure 1

NOSE



Loop Element Diagram

GE SIGNA 1.5T 8 Channel High Resolution Head Array Coil Block Diagram



SECTION 2 – SETUP AND CALIBRATION

2-1 Coil Installation

2-1-1 Special Install Notes

None

2-1-2 Installing the Coil

The name for this coil is: **8HRBRAIN**

Add the coil using the Configuration File Manager. Refer to: Service Methods CD; System Level Procedures; Software Utilities.

2-2 Installation Functional Checks

1. Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.
2. Perform Section 3 – Coil Imaging Performance Verification.

2-3 Periodic Quality Assurance Check

QA checks should be performed during installation and during planned maintenance. Perform the quality assurance checks as outlined below to ensure the coil is operating properly.

1. Check external cable for cracks or cuts.
2. Perform Section 3 – Coil Imaging Performance Verification and record data values in Data Sheet.

SECTION 3 – FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

3-2-1 Tools Required

TOOLS REQUIRED – TABLE 3-2-1

Description	GE Part #	INVIVO #	Qty
18cm Diameter Spherical TLT Phantom	46-265826G6	N/A	1
Phantom Positioner	2317112-3	102558	1

3-2-2 QA Phantom and Coil Setup, and SNR Test Execution

1. Attach the base plate with 8 channel hi res brain coil to the cradle in the same position on the cradle as the standard head coil. Note the serial number of the coil.
2. Slide the coil back to permit access to the head holder as shown in Figure 3-2-2-1.
3. Place phantom ring on head holder as shown in Figure 3-2-2-2. The edge of the ring that goes toward the magnet has edges that overhang the head holder to secure the ring in position.
4. Place head TLT sphere on phantom ring as shown in Figure 3-2-2-3.
5. Slide the coil all the way forward (away from the magnet) on the base plate and set the locks as shown in Figure 3-2-2-4.
6. Establish an axial landmark on the line on top of the coil. Advance to scan.
7. End the previous exam if this has not already been done.

Figure 3-2-2-1



Slide Coil Back to Access Head Holder

Figure 3-2-2-2



Place Phantom Ring on Head Holder

Figure 3-2-2-3



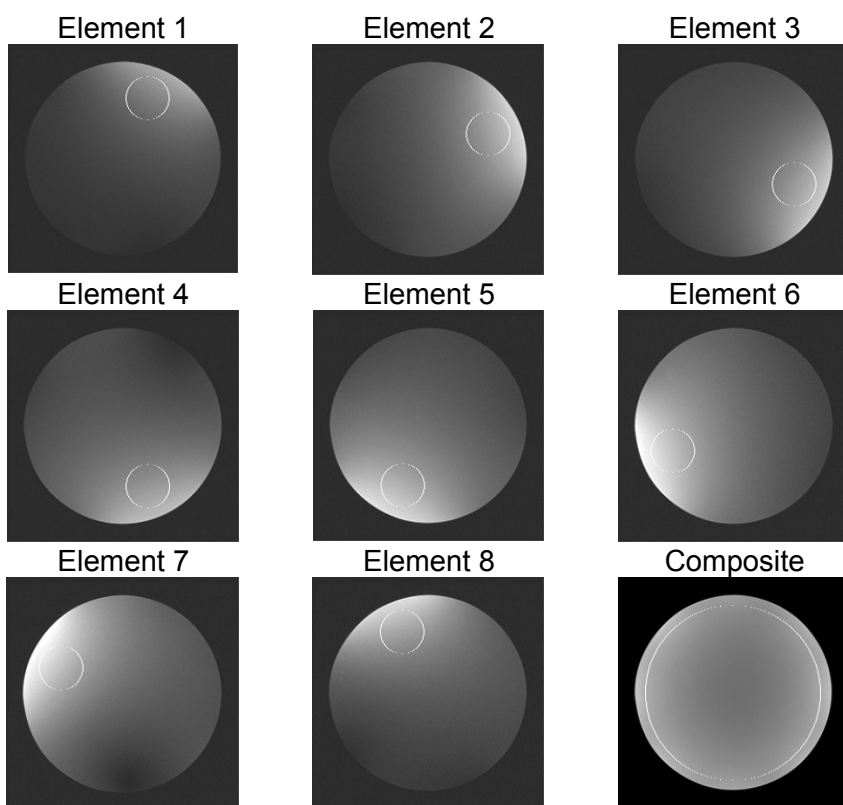
Place Head TLT Sphere on Phantom Ring

Figure 3-2-2-4



Sphere Aligned with Coil

Figure 3-2-2-5



Individual Element ROIs

**Continue for
14x Software Versions or Greater**

8. On the Common Service Desktop, select Image Quality, and then the MCQA tool.
9. Make the proper coil selection and press the "Start Scan" button.

**Continue for 11x and 12x
Software Versions**

8. From the service browser, select Image Quality -> Coil SNR -> [Click here to start this tool](#)
9. Select the coil type you are testing, enter the serial number and select Start Scan.

Note: Additional documentation can be accessed by selecting File -> Help -> 8HRBRAIN

Note: The SNR tool requires that the **8HRBservice** service configuration file be installed.

The image quality check uses two consecutive acquisitions for the same slice. The signal is measured at the center of the phantom with a ROI of about 60% of the phantom size. The noise is measured on the subtracted image of the two consecutive signal images using the same ROI. SNR is calculated by dividing the Signal mean by the noise standard deviation. All measurements are performed on composite images. Interim images are acquired for reference only.

Proceed to 3-2-5 Acceptance Limits

3-2-3 Data Analysis (10.X Only)

1. In the command window enter **snr_for_8hrbrain_QA** and press enter. A screen like that shown in Figure 3-2-3-1 should appear.

Figure 3-2-3-1

Output data path: /usr/g/service/data

Exam number: 50000

Signal series: 1

Noise series number: 2

Coil serial number: none

Calculate SNR

SNR Results:

QA Mode SNR Analysis Screen

2. In the Exam number window enter the exam number as shown in Figures 3-2-3-2 and 3-2-3-3 (The Delete key may be used instead of Backspace).
3. In the Coil serial number window, enter serial number and bay number as shown in Figures 3-2-3-4 and 3-2-3-5 (Optional).

Figure 3-2-3-2

Output data path: /usr/g/service/data

Exam number: 50000

Signal series: 1

Noise series number: 2

Coil serial number: none

Calculate SNR

SNR Results:

Highlight Digits for Exam Number

Figure 3-2-3-3

Output data path: /usr/g/service/data

Exam number: 51897

Signal series: 1

Noise series number: 2

Coil serial number: none

Calculate SNR

SNR Results:

Type In Exam Number

Figure 3-2-3-4

Output data path: /usr/g/service/data

Exam number: 51897

Signal series: 1

Noise series number: 2

Coil serial number: none

Calculate SNR

SNR Results:

Highlight Coil Serial Number

Figure 3-2-3-5

Output data path: /usr/g/service/data

Exam number: 51897

Signal series: 1

Noise series number: 2

Coil serial number: U7514 b3a

Calculate SNR

SNR Results:

Enter Coil Serial Number and Bay Number

4. Click the Calculate SNR button as shown in Figure 3-2-3-6.

Figure 3-2-3-6

Output data path: /usr/g/service/data

Exam number: 51897 Coil serial number: U7514 b3a

Signal series: 1

Noise series number: 2

Calculate SNR

SNR Results:

Click Calculate SNR Button

Figure 3-2-3-7

Data Path: /usr/g/service/data

Exam number: 50399 Coil serial number: noneP4-1 b4a

Signal series number: 1

Noise series number: 2

Calculate SNR

SNR Results:

```
phantom center =(128,129)
Individual Element NEMA SNR from composite images
ROI    signal    noise    snr
1      1384.1    16.07   121.8
2      1449.1    15.97   128.3
3      1514.1    15.43   138.8
4      1524.1    16.09   134.0
5      1613.9    16.77   136.1
6      1682.1    16.51   144.1
7      1610.8    16.58   137.4
8      1478.0    17.05   122.6
mean   1532.0    16.31   132.9
stdev   98.2     0.52    7.9
stdev%   6.4     3.16    6.0
Composite SNR Using NEMA Method
ROI    signal    noise    snr
C      1473.3    16.63   119.8
serial number noneP4-1 b4a
```

- Results will appear as shown in Figure 3-2-3-7.
- Table 3-2-3-1 shows the test results that correspond to each test. For further explanation, refer to Section 3-2-4, Plotting Tool Procedure.

QA Mode SNR Results

EXAMPLE OUTPUT OF TOOL – TABLE 3-2-3-1

phantom center =(131,137)				
Individual Element NEMA SNR from composite images				
ROI	signal	noise	snr	Using ROIs shown in Figure 3-2-2-5 signal is measured on each of the two composite images and averaged. The same ROI for each element is used to measure noise on the one filtered difference image
1	1835.0	19.39	133.8	
2	2054.6	19.61	148.1	
3	2117.0	19.44	154.0	
4	2005.0	19.79	143.3	
5	1994.5	19.49	144.7	
6	2080.0	19.70	149.3	
7	2029.9	20.24	141.8	
8	1848.8	20.19	129.5	
mean	1995.6	19.73	143.1	
stdev	102.8	0.33	8.1	
stdev%	5.2	1.66	5.7	
Composite SNR Using NEMA Method				Composite image results
ROI	signal	noise	snr	
C	1920.1	19.99	129.8	
serial number U7514 bay3a				

Note: The results are also stored in a file in the /usr/g/service/data directory. The filename will be similar to 8hrbrain_51897_c7ga0522.qa where 51897 is the exam number.

3-2-4 Plotting Tool Procedure (10.X Only)

1. Open a command window on the tools desktop.
2. Navigate to the /usr/g/service/cclass directory.
3. Type the command **plot_8hrbrain_data** and press the enter key.
4. The screen shown in Figure 3-2-4-1 will appear.
5. The path to data files, /usr/g/service/data, should be selected by default. If it is not, highlight the text in the type in the correct path.
6. Click on the “List and Select one of below data files” button as shown in Figure 3-2-4-2.
7. Click on the name of the file to be plotted as shown in Figure 3-2-4-3.
8. Click on the Plot Data button as shown in Figure 3-2-4-4 to process the file.
9. Example results for a nominal coil are shown in Figure 3-2-4-5. Coil with poor SNR in Figure 3-2-4-6. Coil with shorted blocking diode in Figure 3-2-4-7. It is interesting to note that the coil whose results are shown in Figure 3-2-4-7 marginally passes the composite image SNR limits but that element 3 performance in individual element results is severely out of normal range.

Figure 3-2-4-1

8HRBRAIN coil Data

Datapath /usr/g/service/data

List and Select one of below data files Plot Data

Enter Datapath

Figure 3-2-4-2

8HRBRAIN coil Data

Datapath /usr/g/service/data

List and Select one of below data files Plot Data

List Files

Figure 3-2-4-3

8HRBRAIN coil Data

Datapath /usr/g/service/data

List and Select one of below data files Plot Data

8hrbrain_52047_c7j02525.qa
8hrbrain_50933_c9qb2540.qa
8hrbrain_50690_c6ta1748.qa

Select File

Figure 3-2-4-4

8HRBRAIN coil Data

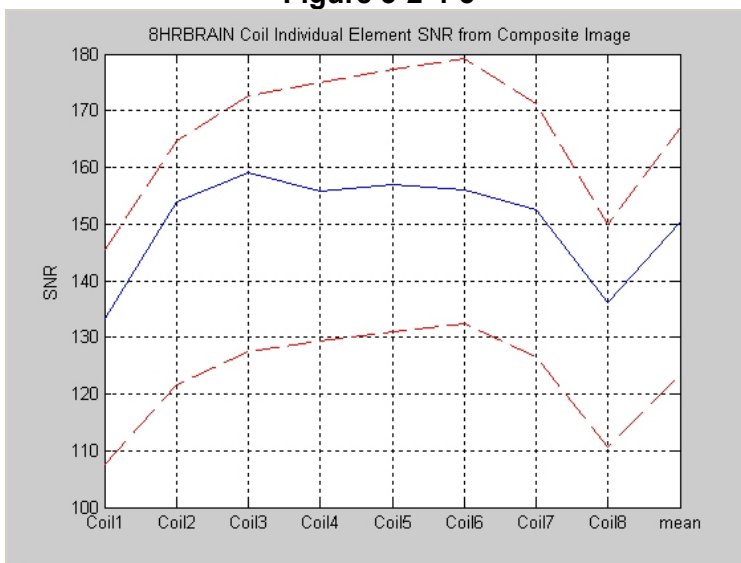
Datapath /usr/g/service/data

List and Select one of below data files Plot Data

8hrbrain_52047_c7j02525.qa
8hrbrain_50933_c9qb2540.qa
8hrbrain_50690_c6ta1748.qa

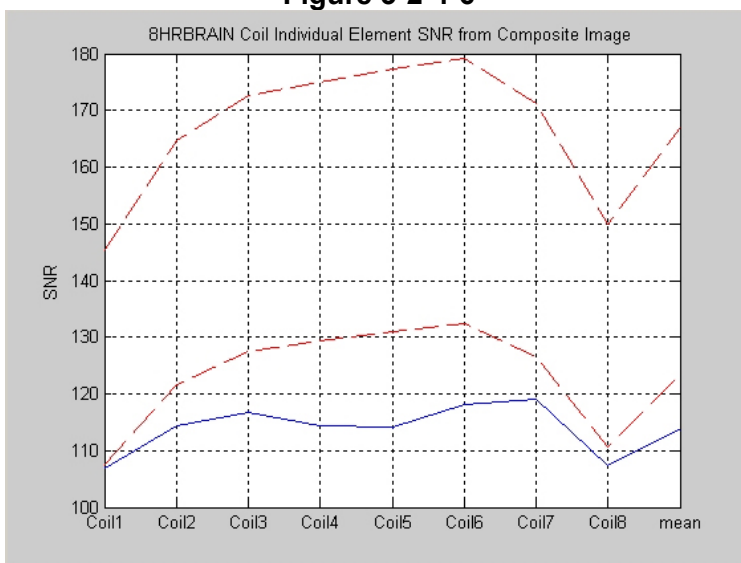
Process File

Figure 3-2-4-5



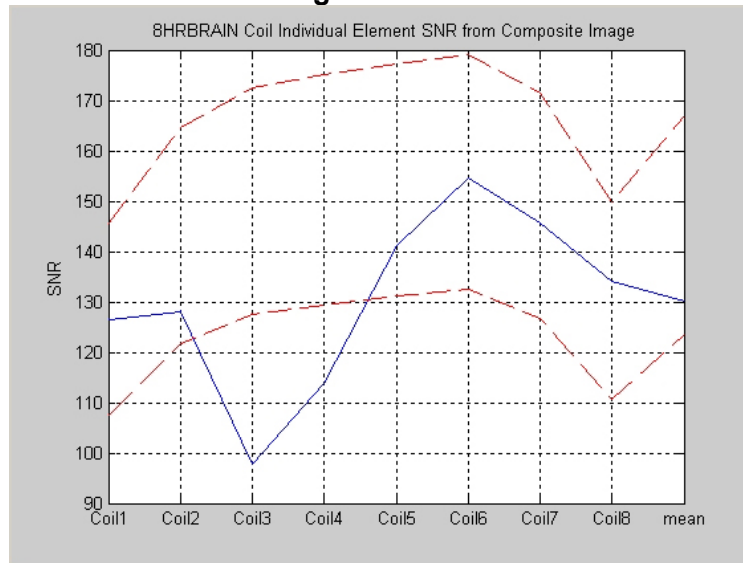
Nominal Individual Element Results

Figure 3-2-4-6



Low SNR Individual Element Results

Figure 3-2-4-7



Shorted Blocking Diode Individual Element Results

3-2-5 Acceptance Limits (10.X and 11.X)

- Record the coil serial number and the associated filename in Table 3-2-5-1 for future reference.

Note: The analysis tool can be used to process multiple runs by entering a new exam number and coil serial number and pressing the Calculate SNR button again. It is not necessary to exit the analysis tool after each run.

- Using Table 3-2-3-1 as a reference, compare the test results to the values in Tables 3-2-5-2 and 3-2-5-3. For additional clarification, refer to Section 3-2-4, Plotting Tool Procedure.

COIL SERIAL NUMBER AND FILENAME – TABLE 3-2-5-1

Date	Coil Serial Number	Filename

COMPOSITE SNR ACCEPTANCE LIMITS – TABLE 3-2-5-2

Nominal	Lower Limit	Upper Limit
131.3	118.2	144.5

INDIVIDUAL ELEMENT SNR LIMITS – TABLE 3-2-5-3

	element	nominal	lower limit	upper limit
individual elements from composite image	1	126.6	107.6	145.6
	2	143.2	121.7	164.6
	3	150.1	127.6	172.6
	4	152.2	129.4	175.1
	5	154.2	131.0	177.3
	6	155.8	132.5	179.2
	7	149.0	126.6	171.3
	8	130.3	110.8	149.9
	mean	145.2	123.4	166.9

3-2-6 Test Explanation and Results Interpretation

The 8 channel hi res brain coil is a moderately complex phased array. The Quality Assurance tool acquires data looking at the composite image (like the customer sees). Individual element performance is evaluated using the composite image.

Signal images are acquired using a FSE-XL sequence with CV saveinter set to 1 to save intermediate images into the image database. The scan is taken twice for a total of 18 images. Next, a no-RF noise scan is taken, also with saveinter on for a total of 9 noise images. Individual element signal and noise images no longer yield results that are displayed to the user.

The test employs two types of image analyses: Separate signal and noise images are evaluated for individual element signal, noise and SNR, much like other automated surface coil tests such as those used for both types of 1.5T CTL coils. In addition, a filtered version of the NEMA SNR measurement method is also used. The filter takes out adverse effects of subtle ghosting on noise measured in difference images. Both types use the same individual element regions of interest (ROIs) which are shown in Figure 3-2-2-5. The Composite SNR using NEMA method also uses the ROIs shown in Figure 3-2-2-5. Only results from composite images are displayed to the user.

- Composite SNR acceptance limits are shown in Table 3-2-3-3.
- Example output of the tool is shown in Table 3-2-3-1.

An automated plotting tool is available to evaluate a particular result file against the individual element acceptance limits shown in Table 3-2-3-4. See Section 3-2-4, Plotting Tool Procedure, for instructions. Example plots from good and defective coils are shown in that section.

3-3 External Cable Check

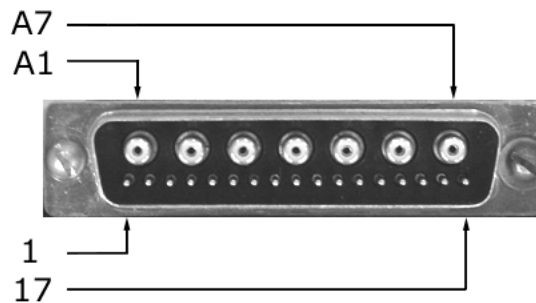
Each MCP signal connection has a DC blocking capacitor in the cable connector. Therefore, continuity of the RF signal conductors cannot be determined in the usual way, by measuring from pin to pin using a multimeter. The external cable check must be performed in two parts, section 3-3-1 for the RF signal conductors and 3-3-2 for the PIN diode control conductors.

Use Figure 3-3 for pin identification while performing the procedures in sections 3-3 and 3-4.

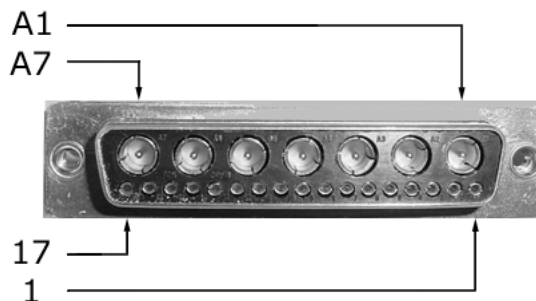
System and Coil Connectors Pin Identification Figure 3-3
Bendix System Connector



SubD Coil Cable Connector



SubD Coil Connector



3-3-1 RF Signal Channels

The following procedure will rule out a short to GND for each of the 16 signal pins detailed in Table 3-3-1. This check does not, however, determine continuity for each of the MCP signal connections.

Check the coil cable using the following procedure. Referring to Section 5-2, External Cable Replacement, disconnect the SubD coil cable connector. Using a digital multimeter, measure each of the 16 signal pins designated in Table 3-3-1 with respect to GND. Verify that the signal pins are not shorted to GND.

EXTERNAL CABLE EXPECTED READINGS – TABLE 3-3-1

Signal Channel	SubD Coil Cable Connector	GND
MCP1	A7	1,3 and shield
MCP2	A6	
MCP3	A5	
MCP4	A4	
MCP5	A3	
MCP6	A2	
MCP7	2	
MCP8	A1	
Signal Channel	Bendix System Connector	GND
MCP1	B1	A12, A14, A16, A18, A20, A22, A24 and A26
MCP2	A2	
MCP3	B3	
MCP4	A4	
MCP5	B34	
MCP6	A35	
MCP7	B36	
MCP8	A37	

If one or more of the pins are shorted to GND, replace the coil cable.

3-3-2 PIN Diode Control Channels

With the coil cable disconnected, use a multimeter to determine continuity between the designated pins for each control channel in Table 3-3-2. Also verify that the control pins are not shorted to GND.

EXTERNAL CABLE EXPECTED READINGS – TABLE 3-3-2

Control Channel	SubD Coil Cable Connector	Bendix System Connector
MC 1	17	B12
MC 2	16	B14
MC 3	15	B16
MC 4	14	B18
MC 5	4	B20
MC 6	5	B22
MC 7	6	B24
MC 8	7	B26
GND	1,3 and shield	A12, A14, A16, A18, A20, A22, A24 and A26

If one or more of the readings are > 3 ohms, replace the coil cable. If all readings are < 3 ohms, proceed to the **Diode Check**.

3-4 PIN Diodes Check

With the coil cable connected to the coil, use a multimeter to measure the diode drop across the coil connector pins detailed in Table 3-4. Use Figure 3-3 for pin identification while performing this procedure.

PIN DIODE EXPECTED READINGS – TABLE 3-4

Bendix System Connector		Reading
Positive Lead	Negative Lead	
B12	GND [A12]	1.7 ± 0.2
GND [A12]	B12	Open
B14	GND [A14]	1.7 ± 0.2
GND [A14]	B14	Open
B16	GND [A16]	1.7 ± 0.2
GND [A16]	B16	Open
B18	GND [A18]	1.7 ± 0.2
GND [A18]	B18	Open
B20	GND [A20]	1.7 ± 0.2
GND [A20]	B20	Open
B22	GND [A22]	1.7 ± 0.2
GND [A22]	B22	Open
B24	GND [A24]	1.7 ± 0.2
GND [A24]	B24	Open
B26	GND [A26]	1.7 ± 0.2
GND [A26]	B26	Open

If one or more of the readings indicate a short, replace the coil. If all of the above readings are correct, redo 3-2 *Coil Imaging Performance Verification*. If the problem still exist, replace the 1.5T 8 Channel High Resolution Brain Array.

3-5 Mechanical Hardware Check

Not Applicable

3-6 Troubleshooting Tips

If the following error is displayed, there may be a problem with the Coil ID coil or system hardware: *The prescribed RF coil and detected RF coil do not match.*

Trouble-shooting instructions are available online by accessing the service browser:

Diagnostics -> Peripherals -> SRI Functional Tests -> Coil ID (click directly on the text).

Start the diagnostic by selecting **SRI Coil ID** and **Run**. The diagnostic will display the contents of the coil ID chip whenever the connector is inserted. Follow the trouble-shooting instructions if the correct ID is not displayed. When the trouble-shooting is complete, and if the correct ID is still not displayed, replace the cable.

SECTION 4 – MAINTENANCE

4-1 Coil Care



Detach coil connector from scanner before attempting to clean. Do not reattach after cleaning until coil has dried completely. Having the coil attached to the system during cleaning or when it is wet may result in electrical shock.



Do not spray or pour cleaning solution directly on coil. Do not submerge coil in solution. The coil contains sensitive electronics components that could be damaged by the solution.

The GE SIGNA 1.5T 8 Channel High Resolution Brain Array, Baseplate and Head Pads may be cleaned by wiping with a cloth dampened with a solution of 30% isopropyl alcohol and 70% tap water or a 10% Bleach solution.

4-2 Special Care Requirements

None

SECTION 5 – REPLACEMENT

Simple removals that are clearly obvious are not described here.

Unless otherwise noted, the steps for re-assembly are simply the reverse order of the steps described for disassembly.

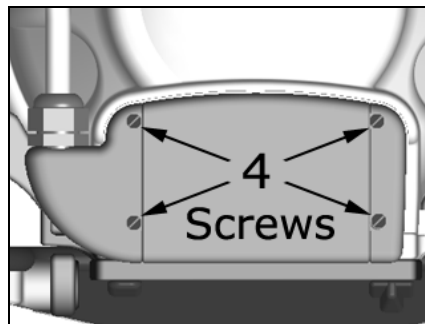
5-1 Disassembly of Coil

Not Applicable

5-2 External Cable Replacement

The 4 coil connector mounting screws are located at the superior end of the coil. Remove the 4 coil connector mounting screws as detailed in Figure 5-2-1. Remove the coil connector by pulling away from the coil in the superior direction.

Figure 5-2-1



Remove 4 Screws



Cables bearing this Coil ID label are compatible with software versions 10.X and newer. With software versions 11.X and newer, the system will generate an error when an incompatible coil is prescribed.

5-3 Mechanical Hardware Replacement

Instructions for FRUs containing mechanical hardware replacement parts are included with the applicable FRU.

SECTION 6 – RENEWAL PARTS

6-1 Field Replaceable Units

FIELD REPLACEABLE UNITS LIST – TABLE 6-1

Description	GE Part #	INVIVO #
1.5T 8 Ch HiRes Brain Array / Baseplate Assm.	2317112-2	101844
Cable	2317112-8	101853
Phantom Positioner	2317112-3	102558
1.5T 8Ch HiRes Brain / Latch and Coil Adapter Assy.	2317112-12	104537
Mirror Kit	5146634-7	4535-302-9076*
Latch Kit	2317112-13	4535-302-9077*
Head Bucket Kit	5147134-5	4535-302-9078*

* The last digit of the Invivo Part number is the part's rev #.



Cables bearing this Coil ID label are compatible with software versions 10.X and newer. With software versions 11.X and newer, the system will generate an error when an incompatible coil is prescribed.

6-2 Other Replaceable Accessories

OTHER REPLACEABLE ACCESSORIES LIST – TABLE 6-2

Description	GE Part #	INVIVO #
Thick Head Pad	2317112-9	102985
Thin Head Pad	2317112-10	102984
Spacer Pad	2317112-7	101622
Wedge Pad	2317115-8	101407

SECTION 7 – APPENDIX

7-1 Schematic

Refer to Block Diagram under Section 1-5 Theory of Operation.

7-2 Coil Configuration

Check the system coil configuration file for the 8HRBRAIN coil configuration. If 8HRBRAIN has not been installed, select 8HRBRAIN from the list of available coils, and using the config file manager, install the following coil names: 8HRBrain and 8HRBService.

Revision History

Rev	Date	Author	Primary Reason for Change
A	11-20-01	L. Hyler	Initial Release
B	04-15-02	L. Hyler	Completed External Cable Check & PIN Diode Check
C	06-12-02	L. Hyler	Complete Functional Checks, Section 3
D	07-29-02	L. Hyler	Make Corrections Outlined in GE Formal Review Log 07-26-02
			Add Automated QA Section
E	08-13-02	L. Hyler	Increased Length of Table 3-2-3-2
F	08-20-02	L. Hyler	Corrected Minor Typos
01	08-23-02	L. Hyler	Initial Issue
01A	12-05-02	L. Hyler	Update QA Procedure per GE
01B	01-09-03	L. Hyler	Replace Figures 3-2-4-5, 3-2-4-6, 3-2-4-7
01C	01-09-03	L. Hyler	Add text back to Section 3-2-4 item 9
02	01-15-03	L. Hyler	Release changes – 01A, 01B, 01C
03	05-19-03	L. Hyler	Replace Head Pad Part Numbers with Thin Head Pad and Thick Head Pad Part Numbers
04	06-18-03	L. Hyler	New Baseplate Illustrations
			Section 3-2-2 correct path to /usr/local/bin/CoilsTools
05	08-11-03	L. Hyler	Changed Table 3-4 spec. from 2.0 ± 0.2 to 1.7 ± 0.2
06	11-04-03	L. Hyler	Redesigned section 3-2-2 for both version 10.X and 11.X s/w Add Coil ID Information to Sections 3-6, 5-2, and 6-1
07	04-02-04	L. Hyler	Added Latch and Coil Adapter Assy., Section 6-1
08	07-06-05	D. Visscher	Rebrand MRI Devices with Invivo (DCN 1374)
09	6/19/08	V. Thacker	Customer Change Request CCN0021 (DCN3994)
10	6/24/09	R.M. Scott	Add Coil Configuration (DCN 5847)
11	9-29-10	K. Hilton	Removed hyper link
12	09-24-12	L. Hyler	Updated Language Policy For Service Documentation, added new lower level FRUs and updated section 5-3. (DCN11910)